Dika Muhammad Azis

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PERSONAL

Final-year Engineering Physics student specializing in industrial automation, control systems, and mechanical-electrical integration. Driven by a clear vision to become an advanced engineer and committed to long-term professional growth through hands-on industrial problem-solving. Experienced in PLC programming, circuit design, and developing efficiency-focused automation systems. Seeking a full-time Automation Development Engineering role to leverage expertise in PLC, IoT, and AI-driven control algorithms in designing robust, future-ready industrial solutions

EDUCATION

Bandung Institute of Technology (ITB)
Bandung
B.Eng. in Engineering Physics (Instrumentation and Control) *Aug. 2022 - now (expected graduation : Aug. 2026)*

- Current GPA: 3.68/4.0
- Related coursework :
    - Signal Processing
    - Electric Circuits and Electronics
    - Logic and Digital Systems
    - Machine Learning for Control
    - Biomedical Instrumentation and Control
    - Multivariable Control System
    - Industrial Internet of Things (IIoT)
    - Robotics and Automation
    - Instrumentation and System Integration in Industry
- Thesis : Road Damage-Aware Motion Planning via Deep Reinforcement Learning for Enhanced Comfort and Safety in Autonomous Vehicle.
- Scholarships : Karya Salemba Empat (KSE) Foundation Scholarship (*Dec. 2023 – Aug. 2024*), Program Indonesia Pintar (PIP) Scholarship (*2025*).

PROFESSIONAL EXPERIENCE

Research Assistant Jan. 2026 - July. 2026
Directorate of Research and Innovation , ITB *Bandung*

- Conducted research on Road Damage-Aware Motion Planning via Deep Reinforcement Learning for Enhanced Comfort and Safety in Autonomous Vehicle
- Prepared and submitted a manuscript based on this research to a Q2-indexed international journal

Industrial Automation Project Engineer Dec. 2025 - Jan. 2026
Industrial Instrumentation and Integration Laboratory, ITB *Bandung*

- Integrated custom build Automatic Test Equipment and Mitsubishi PLC with OPC servers, SQL Databases, Ignition SCADA, and MATLAB.
- Configured PLC I/O mapping for Digital (LEDs, switches) and Analog (7-segment, potentiometers) signals.
- Authored technical documentation and teaching modules to facilitate hands-on learning.

System Engineer Intern July 2025 - Aug. 2025
PT AVS Simulator *Bandung*

- Conducted theoretical torque analysis of the MOZA R12 direct-drive system using iRacing WheelCheck.
- Designed electrical diagrams and 3D CAD custom steering utilizing JTEKT Electronic Power Steering.
- Developed equations to convert telemetry parameters from MOZA Pit House software.

Embedded Engineer Intern June 2024 - Aug. 2024
Cultination *Bandung*

- Developed an automated fish pond monitoring system integrated with smartphones across 3 ponds, tracking real-time parameters including temperature, water level, weather, and air flow.
- Engineered a smartphone monitoring and control system for a three-phase motor utilizing Mitsubishi FR-D700 inverter, regulating on/off states, speed, and monitoring voltage inputs/outputs.

RESEARCH AND PUBLICATION

Pothole Avoidance and Motion Planning for Autonomous Vehicles using Deep Reinforcement Learning within the CARLA Simulator
Bandung Institute of Technology Instrumentation, Control and Automation Research Group
15th Asian Control Conference (ASCC) 2026
Status : Accepted, Oral Presentation

- Engineered an autonomous motion planning control system utilizing Deep Reinforcement Learning within the CARLA simulator to achieve dynamic, real-time pothole avoidance.
- Developed AI algorithms and simulation environments using Python and C++ to process sensor data and optimize vehicle navigation accuracy.
- Detailed the control system performance and collision-avoidance metrics.

Fault Detection of Mining Truck Tire Using a Switched Fuzzy Interval Observer
15th Asian Control Conference (ASCC) 2026
Status : Accepted, Poster Presentation

- Engineered a cost-effective, model-based fault detection strategy for mining truck tires utilizing a Switched Takagi-Sugeno (T-S) fuzzy framework and an Interval Observer.
- Detailed the mathematical modeling and algorithm performance, demonstrating the observer's capability to effectively distinguish between healthy and faulty tire conditions.

ACADEMICS PROJECTS & COMPETITION

IIoT 3-Axis Automated Storage and Retrieval System | *PLC, MQTT, HMI, Postgres* Oct 2025 - Jan. 2026

- Developed the control logic for a 3-axis automated pick-and-place storage system utilizing an Omron PLC.
- Designed an interactive Human-Machine Interface (HMI) utilizing Weintek to provide real-time operational visualization, system diagnostics, and on-site manual control.
- Engineered a secure Industrial IoT (IIoT) cloud computing architecture using the MQTT protocol and Node-RED, enabling remote monitoring and secure network integration.

Techno Soccer Robot Competition | *Arduino IDE, Solidworks, Eagle*, *Android Studio* Oct. 2025

- Designed custom PCB and integrated motion control hardware for the soccer robot.
- Developed motion control algorithms and a Human-Machine Interface for system operation.
- Ranked Top 16 out of 50+ participating teams.

ORGANIZATIONAL EXPERIENCE

Vice Director Mechanical Division July 2024 – June 2025
ITB Robotics Unit, Traditional Dance Robot Team *Bandung*

- Co-led a team of 5 people, managing end-to-end mechanical development of two humanoid robots.
- Designed humanoid robot components using SolidWorks and manufactured parts utilizing 3D printing.
- Procured mechanical materials and assembled 3D-printed parts to construct fully functional robot hardware.

TECHNICAL SKILLS

Certification : Certified SOLIDWORKS Associate (CSWA).
Developer Tools : MATLAB, EAGLE, LTSpice, SolidWorks, Android Studio, Arduino IDE, Ignition, Node-RED, VS Code, OPC KepServer, GX Works 2, GX Works 3, Twido Suite, sysML, Git.